LabelLens: Seeing a Bluesky Post Through Other People's Moderation

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ABSTRACT

On Bluesky, moderation is *stackable*: anyone can run a labeling service, and each user chooses which of these *labelers* to trust. Two people can therefore see the same post very differently, and nobody, including its author, sees the whole picture. We present LabelLens, a serverless web tool that asks every public labeler about a post or an account at once and renders the item through several subscription sets with the official client's moderation code, side by side. A companion page reports a network-wide measurement of all 576 declared labelers. Only 204 of their 496 published endpoints answer queries, and 65% of those allow a browser to read the answer. In a uniform sample of labelled posts, the twenty most-liked protective labelers change how 54.1% are presented to a subscriber compared with a logged-out visitor, and 18.6 pp more are hidden or put behind a warning, mostly because clients apply an author's account labels to their posts. 95.2% of the changed posts carry a label from a single service; without it the change is 6.8%. The twenty most-liked labelers overall change 32.1%. LabelLens runs from a link, with no login and no server.

CCS CONCEPTS

• **Information systems** → **Social networking sites**; • **Human-centered computing** → *Collaborative and social computing systems and tools.*

KEYWORDS

decentralized social media, Bluesky, AT Protocol, content moderation, labeling, transparency

1 INTRODUCTION

Bluesky replaces a single platform moderator with *labels* [4, 9]. A labeler is an account that signs short values such as `spam` or `ai-imagery` and attaches them to posts or accounts [1]. Every client applies Bluesky's own labeler, and a signed-in user may subscribe to up to twenty more and decide, per label, whether to hide, warn or ignore. This realises the long-standing idea of moderation *middleware* [8], but it also fragments what people see. An author cannot tell which labels follow their account, a reader cannot tell what other readers are shown, and researchers have so far studied either the network's infrastructure [2] or Bluesky's own labeler [11], which explicitly leaves third-party labelers to future work.

LabelLens makes this fragmentation visible for one item at a time. Given a post link or a handle, it (i) runs a *census*, asking every labeler in a network-wide directory what it says about the post, its author and the author's profile, and (ii) shows the item through *lenses*, that is, subscription sets evaluated with the moderation code of the official client library. A companion *Network* page summarises a measurement of all labelers. The demonstration offers three things: a tool anyone can run from a link, a census method that copes with a network where most services are unreachable from a browser, and measured evidence of how far the stacks diverge. The demo belongs to the Social Networks and Social Media track (decentralized and federated platforms) and the Responsible Web track (content moderation and platform governance).

2 LABELLENS

LabelLens is a static site (GitHub Pages) of about 610 lines of JavaScript; every request goes from the visitor's browser to public endpoints. Figure 1 shows the Inspect view.

Directory. Labelers are ordinary accounts that publish an `app.bsky.labeler.service` record and a `#atproto_labeler` endpoint in their DID document [1]. We list the accounts holding such a record through the relay's `listReposByCollection`, resolve each DID document, and ship the resulting directory with the site, leaving out 23 declared endpoints that are not service URLs (e.g. `localhost` or a web page); one script run repeats the whole measurement.

Census. For an item, the browser sends one `queryLabels` request per labeler (32 in parallel) for the post, the author's DID (account labels) and the profile record (only the last two for an account). Each answer is folded into the labels in force now: the latest label per (labeler, subject, value) wins, a negation retracts it, and an expired label is dropped. Only labels whose source is the queried labeler are kept (signatures are not verified), because some services replay other labelers' labels. Many labelers answer but send no CORS header, so a page cannot read them. For the 64 services without CORS whose label streams we collected, a snapshot of their labels in force at collection time, sharded into 4,096 static files by a hash of the subject's DID, fills the gap, and those rows are marked as snapshot data. The public site ships the snapshot only for its example accounts, so it republishes no labels about individuals, and its build refuses any snapshot that is not restricted to them. To show the lenses quickly, the census runs in two phases: first the services that answered at our last measurement (4 s timeout); then those that were offline (4 s) and those that were slow (8 s) are retried, and the lenses are recomputed if any of them answers; a service that answers *too many requests* is not asked again. Because every query reveals the visitor's IP address to the labeler, the page says so before the first query and offers a *privacy mode* that asks only Bluesky's labeler and reads the snapshot (Section 6).

Lenses. A lens is a subscription set. LabelLens passes the census result and the labelers' own label definitions, fetched live, to

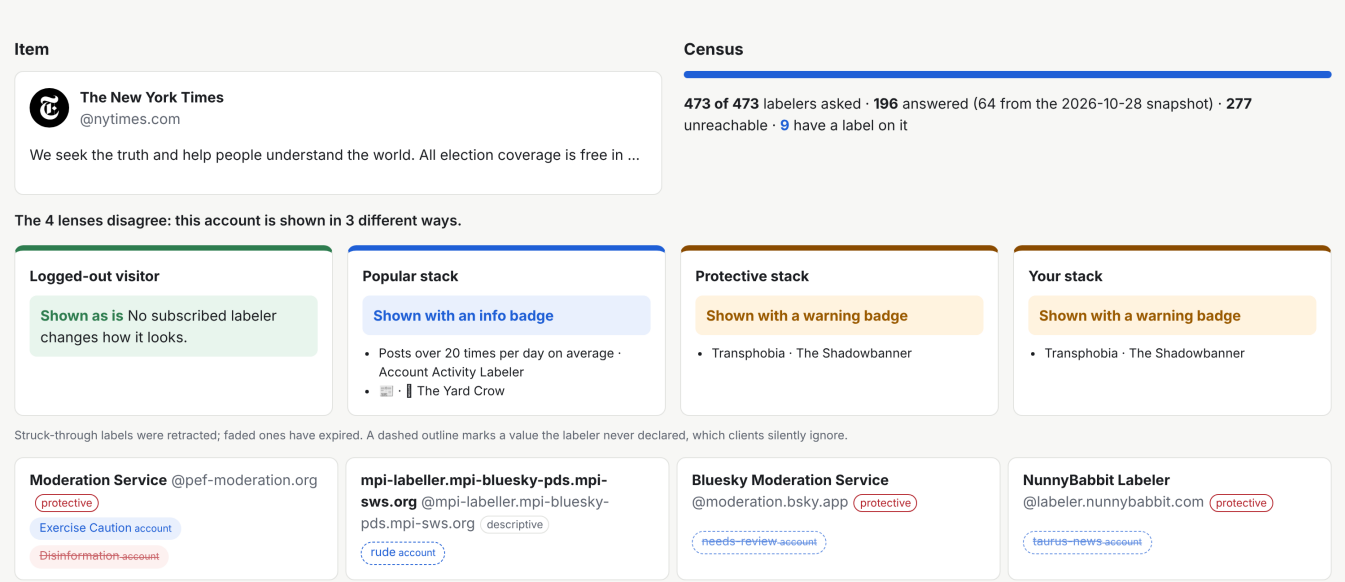


Figure 1: Inspecting a news organisation's account (2026-10-28): 9 labelers label it, and the four lenses show it in 3 different ways. In this run 196 services answered (132 live from the browser, 64 from the snapshot), against 204 endpoints that answered our server-side probe (Section 3). The bottom row keeps only the labelers with a retracted (struck-through) or undeclared (dashed) label. Labels are shown as issued by their labelers, including hostile ones; LabelLens does not endorse them.

`moderatePost` or `moderateProfile` from `@atproto/api`, with default preferences and adult content off, and reports the decision for a feed or a profile list: hidden, behind a warning, warning badge, info badge, or shown as is, with the labels that caused it. The four lenses are a *logged-out visitor* (Bluesky's labeler only, which is also all the public read API returns), a *popular stack* (the twenty most-liked labelers), a *protective stack* (the twenty most-liked labelers that declare a label that hides, blurs or warns) and *your stack*, which starts from the labelers that labelled the item (at most twenty, the client's limit) and can be edited. The two fixed stacks do not stand for a typical user's feed: they are reference points filled to the client's limit of twenty, a stress test of how far subscriptions can move an item, while *your stack* shows a concrete choice.

Network page. The measurement of Section 3 is published as a sortable table, summary tiles and a label-volume chart (Figure 2): reachability, label volume, retractions, labels without effect and time to label, per service and in aggregate only.

3 MEASUREMENT AND EVALUATION

Data. On 2026-10-28, from a cloud workspace, we listed the labeler directory, probed every endpoint once, and replayed each labeler's label stream with `subscribeLabels` from sequence zero. A stream that stopped before the sequence number seen live at its start was resumed from its last event (7 services needed this). For the 8 services whose stream had passed sequence number three million, only the last three million were replayed (Table 2). Replay reached the live head for 34 services and stopped short of it for 1; for 144 quiet services that emitted no live event, completeness cannot be checked. Because the logged-out lens uses Bluesky's labeler alone, its row in Table 1 reflects that labeler's window, which starts on 2026-09-10. A run from a GitHub-hosted runner on 2026-10-21 gave similar reachability (199 answering, 129 browser-readable), so the failures below are the services', not the vantage point's. Generative AI use is described in the acknowledgments.

Reachability. Of 576 accounts that declare a labeler, 496 publish an endpoint, of which 23 are not valid service URLs (e.g. `localhost` or a web page). Only 204 (41% of published endpoints) answer a `queryLabels` request; among the valid ones, the rest cannot be reached (48), fail TLS (33) or answer with an HTTP error or a body that is not JSON (110). Of the answering services, 132 (65%) send a CORS header that admits the site's origin. A browser-only client such as a third-party app or LabelLens itself therefore sees at most 27% of declared endpoints, which is why the census needs its snapshot fallback. Labeler operators can fix this with one response header, and an AppView query for third-party labels would remove the need for per-labeler fan-out.

Labels. We collected 72.2 million labels from 160 services (72% on posts, 27% on accounts). Volume is concentrated: the five largest services issued 69% of all labels. 4.0% of labels carry a value their labeler never declared; clients drop such labels silently, and for 18 services they are the majority. 5.5% of (subject, value) pairs end retracted. Protective labelers differ widely in speed: among the 24 with at least 100 post labels, the median time from post creation to label is under ten seconds for 11 (automated classifiers) and over an hour for 9.

Overlap and divergence. Stacks rarely agree because they rarely overlap: of the 0.6 million accounts labelled by any protective labeler, 10.2% are labelled by two or more. To measure what this means for readers, we drew a uniform sample of labelled subjects (a subject is kept when its hash is 0 modulo 1,000; 40,084 posts

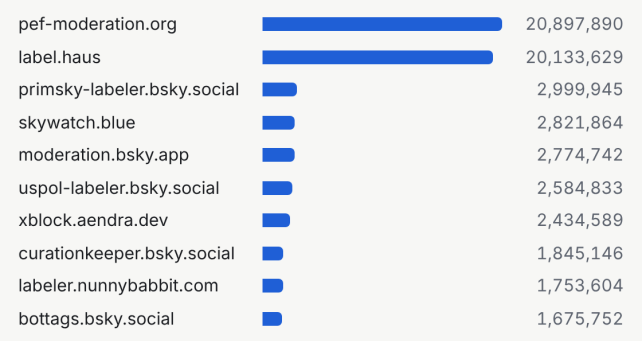


Figure 2: Label volume of the ten largest services on the Network page (2026-10-28). Services whose stream had passed sequence number three million, such as `moderation.bsky.app`, were replayed from a recent window, so their bars count that window only.

and 2,263 accounts; a post carries its author's account labels) and evaluated each lens offline with the same moderation code as the site (Table 1). For posts, the protective stack changes the presentation of 54.1% relative to a logged-out visitor: 23.1% are hidden instead of 5.8%, and 18.6 pp more are hidden or carry a warning. Most of this comes from accounts, since clients apply an author's account labels to each of their posts: judged on their own labels alone, the protective stack changes 45.9% of posts (1.9 pp more hidden or warned), mostly with info badges. The popular stack changes 32.1% (1.0% on the posts' own labels), almost all with info badges: the most-liked labelers are mostly descriptive (badges, interests, verification), so popularity is a poor guide to protection. The effect is concentrated: 95.2% of the changed posts carry a label from one service, `pef-moderation.org`, which flags links to other platforms and hides by default the posts of accounts it labels as spreading disinformation or abusive. Without it the protective stack changes 6.8% and hides 7.9% (7.0% changed when the next most-liked protective labeler takes its place), while leaving out any other single service gives 53.4–54.1%. Which labelers a user adds matters more than how many, and this is exactly what the census and the editable stack expose for a single item. Subscribing to the labelers that labelled the item (at most twenty) changes 88.8% of posts.

Census performance and accuracy. We timed the census on 20 items from the sample (another 11 could no longer be loaded: deleted, suspended or hidden) in headless Chromium from our workspace. On the public site, the lenses appeared 4.2 s after the click at the median (6.6 s at most; the first census pass took 3.6 s), with 132 labelers answering live for every item; the recheck of silent services finished after 15.6 s and recovered at most 1 more per item. A local build that loads the full snapshot, which we do not publish, raises the median to 194 answering services. Against the labels in the collected streams, the census on the public site reproduced every label for 44% of 200 sampled items (80% of labels). Every difference came from services without CORS, whose labels the public snapshot leaves out (89 items), or were older labels that the census found but our windowed collection missed (52 items); no service was offline and no label had been issued or retracted since collection.

Table 1: How labelled items are shown under each lens: percent of items in a uniform 0.1% sample of the subjects in the collected streams; Δ is the change in hidden or warned items against the logged-out row, in points. Rows are constructed subscription sets, not observed user subscriptions.

Lens	Hidden	Warned	Info	As is	Δ
Posts (n=40,084)					
Logged-out visitor	5.8	0.0	0.0	94.2	–
Popular stack	5.8	0.4	31.6	62.1	+0.4
Protective stack	23.1	1.3	35.5	40.1	+18.6
w/o top labeler	7.9	2.0	2.7	87.4	+4.2
All that labelled it	29.0	13.2	52.4	5.4	+36.4
Accounts (n=2,263)					
Logged-out visitor	0.1	0.0	0.0	99.9	–
Popular stack	0.1	0.5	13.4	85.9	+0.5
Protective stack	2.1	1.1	15.9	80.8	+3.2
w/o top labeler	0.5	0.8	10.2	88.5	+1.2
All that labelled it	6.7	1.4	77.9	14.0	+8.0

Table 2: Services replayed over their last three million sequence numbers: earliest label collected, labels collected, and whether replay reached the live head.

Service	From	Labels (M)	Complete
`primsky-labeler.bsky.social`	2026-10-06	3.0	yes
`skywatch.blue`	2026-01-04	2.8	yes
`moderation.bsky.app`	2026-09-10	2.8	yes
`xblock.aendra.dev`	2025-12-19	2.4	yes
`atls.city`	2026-10-25	1.4	yes
`profile-labels.bossett.social`	2024-12-12	1.2	yes
`papercut.rodeo`	2026-10-26	0.2	yes
`labeler.antisubstack.fyi`	2026-03-16	0.1	yes

Limitations. Lenses are constructed subscription sets: real subscriptions are kept in users' private preferences, and likes are only a proxy for them. Lenses use default per-label preferences, while users may change them, mute words or block accounts. The sample conditions on subjects that some collected labeler labelled, and windowed services contribute only recent labels. The offline evaluation omits authors' self-labels, which live in post records rather than label streams. LabelLens does not verify label signatures. Snapshot data ages from its collection date, and all measurements come from one run on 2026-10-28.

4 DEMONSTRATION

Visitors open LabelLens at https://seungbum.github.io/labellens/ (no login). A typical walk-through takes three minutes. (1) Paste a post link or a handle, or click an example; the census bar fills as 473 services are asked. (2) Compare the four lens cards: the same item may be shown as is to a logged-out visitor, with an info badge in the popular stack and with a warning in the protective stack, each with the labels that caused it. (3) Open *your stack* and toggle labelers to see which one decides the outcome. (4) Scan the per-labeler list, where struck-through labels were retracted, faded ones expired, and dashed ones have no effect. (5) Switch to the Network page to

see how few services are reachable and how unevenly labels are spread. At the conference we run the same public build, on our laptop or the attendee's phone: attendees can inspect their own accounts, and for any account other than the examples, services without CORS are shown as *does not allow browser queries*, so no collected labels about individuals are displayed. This is not a gap in the tool but the point of the census bar: attendees see on their own account the limit measured in Section 3, where only 27% of declared endpoints can be read from a browser. Researchers can rebuild the full snapshot on their own machine with one command from the repository (`make measure derived local`, several hours of collection); that build is never published. The demo needs only a browser and Internet access; if the venue network blocks labeler services, the census still shows what remains reachable, and the recorded video covers the full walk-through. A two-minute video is linked from the site's About page and the repository, https://github.com/seungbum/labellens.

5 RELATED WORK

Bluesky's architecture and composable moderation are described by its developers [4, 9]. Balduf et al. measured the network's components, including a first look at labelers [2]; Singh et al. audited Bluesky's own moderation service [11], and Bono et al. studied blocking as self-moderation [5]. On Mastodon, moderation surfaces as content warnings and instance policies [13]. Transparency work on centralised platforms relies on self-reported statements of reasons in the DSA Transparency Database [6, 12], and the middleware literature argues for third-party moderation services [7, 8]. Community tools list labelers with their endpoints [3] or query one chosen labeler about one subject, as the repository browser PDSls does [10]; LabelLens instead asks every labeler at once, renders the result for constructed subscription sets with the official moderation code, needs no server, and ships a network measurement. It complements these measurement studies with an interactive, per-item view across all third-party labelers, and shows what the middleware model looks like to readers in practice.

6 ETHICAL USE OF DATA

LabelLens reads only public data through public APIs and stores nothing about visitors. Queries go from the visitor's browser to Bluesky and the labelers, so each queried labeler learns the visitor's IP address and the inspected item; official clients avoid this because Bluesky's servers attach the labels a user subscribes to (the `atproto-accept-labelers` header [1]). LabelLens states this before the first query and offers a privacy mode that queries only Bluesky's labeler. The measurement replayed public label streams through each service's documented endpoint, one connection per service (twelve parallel ranges for the largest one), with an identifying user agent. We publish aggregate figures per labeling service only: no list of labelled accounts or posts, and the per-labeler table omits label values, which can describe a person (for example pronouns or political affiliation). For services without CORS, the public site ships collected labels only for its organisational example accounts, its build and a repository check refuse any other snapshot, and the venue demo runs this public build. The raw label logs were deleted after analysis. Labels can be wrong or hostile; showing one does not endorse it, and the interface says so. Examples in the interface and this paper use organisational accounts. No human participants were involved.

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